AAC for Societal Good

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Abstract

Augmentative and Alternative Communication (AAC) systems are essential tools for individuals with speech impairments, yet existing solutions often lack personalization and emotional resonance. This project proposes a hybrid approach that combines fine-tuned Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG) to improve the expressiveness, empathy, and contextual relevance of AAC-mediated conversations. By integrating two emotion-rich datasets Empathetic Dialogues and GoEmotions we enable the system to retrieve emotionally grounded experiences and generate personalized responses aligned with the user's communication style. Experimental results, both automatic and human-evaluated, show significant improvements in fluency, emotional alignment, and personalization. This work demonstrates the potential of LLMs for societal good, offering a promising direction for more human-centered AAC systems.

1. Introduction/Problem Statement

Individuals with speech impairments often rely on Augmentative and Alternative Communication (AAC) systems to express their thoughts, needs, and emotions. However, most existing AAC tools generate generic, templated responses that fail to reflect the unique personality, emotional state, or lived experience of the user. This lack of personalization can result in disconnection, reduced expressiveness, and an inability to participate in meaningful social interactions. While recent advances in Large Language Models (LLMs) have enabled more natural and coherent dialogue systems, these models are typically not tailored to the nuanced needs of AAC users. Moreover, LLMs often hallucinate or produce contextually irrelevant responses, especially when grounded context is missing.

This project addresses the challenge of enhancing AAC communication by integrating LLMs with Retrieval-Augmented Generation (RAG). Our goal is to create an emotionally intelligent conversational agent that not only understands and expresses nuanced emotions, but also retrieves relevant autobiographical or contextual memories to personalize its output. This approach aims to support more empathetic, expressive, and human-like communication for AAC users.

2. Datasets Used

To enable emotionally resonant and contextually grounded dialogue generation, we utilize two publicly available, emotion-rich datasets: EmpatheticDialogues and GoEmotions. Each contributes complementary strengths to our system: one focuses on empathetic multi-turn conversa-

tions, while the other offers fine-grained emotional annotation in single-turn utterances.

2.1 EmpatheticDialogues

The EmpatheticDialogues dataset consists of approximately 25,000 open-domain, multi-turn conversations grounded in emotional situations. Each dialogue is initiated by a speaker describing a real-life situation tied to one of 32 predefined emotions.

2.2 GoEmotions

GoEmotions is a large-scale, multi-label emotion classification dataset comprising 58,000+ Reddit comments. Each comment is annotated with one or more of 27 emotion categories (such as anger, surprise, realization, or love), plus a neutral label. The annotations reflect fine-grained, real-world emotional nuance in diverse, informal language.

3. Methodology / Solution Architecture

Our proposed system enhances AAC-mediated conversations by combining a fine-tuned Large Language Model (LLM) with a Retrieval-Augmented Generation (RAG) pipeline. This architecture is designed to generate responses that are emotionally aligned, contextually grounded, and personalized to an individual user's communication preferences.

3.1 Fine-Tuned LLM

We begin with a pre-trained encoder-decoder language model **T5-Small** from Hugging Face and fine-tune it on a hybrid dataset composed of EmpatheticDialogues and GoEmotions. The training data is formatted to include emotional context `[emotion]`, situational prompts, and expected responses. We used LoRA (Low-Rank Adaptation) to fine-tune efficiently.

3.2 Retrieval-Augmented Generation (RAG)

To personalize responses and minimize hallucinations, we implement a RAG module using **FAISS** and the `all-MiniLM-L6-v2` embedder from SentenceTransformers. This allows us to create a personal narrative database and retrieve contextually relevant emotional experiences.

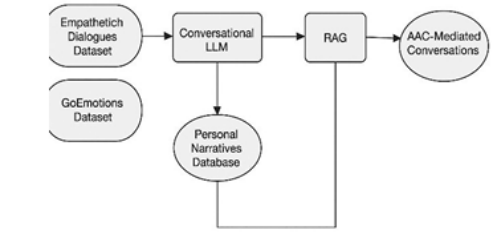

Figure 1: System Architecture: Integration of LLM, RAG, and AAC Interface

4. Experiments and Results

4.1 Baseline: Zero-Shot T5

We tested 't5-small' in a zero-shot setting without fine-tuning. Prompts like `"Respond empathetically: I just lost my job."` yielded fluent but emotionally generic responses such as "That must be tough" or "I'm sorry to hear that."

4.2 Fine-Tuned Model + RAG

Using LoRA, we fine-tuned 't5-small' with a subset of EmpatheticDialogues. We also integrated the FAISS-based RAG retriever powered by 'all-MiniLM-L6-v2'. Sample responses became warmer, more personalized:

- **Sad** – "My dog passed away." → "I'm so sorry. Losing a pet is like losing a best friend."
- **Joyful** – "I got promoted today." → "That's amazing! Congratulations—you totally earned it."

4.3 Automatic Evaluation Results

We evaluated on a 500-sample subset using several NLP metrics. The table below summarizes the scores:

Metric	Score
BLEU	0.0081
ROUGE-1	0.1183
ROUGE-2	0.0138
ROUGE-L	0.1086
METEOR	0.0918
BERTScore (P/R/F1)	0.8658 / 0.8451 / 0.8552

Table 1: Automatic Evaluation Metrics on 500 Sample Subset

4.4 Human Evaluation Results

We used OpenAI's human evaluation API on 50 examples. The table shows average scores (out of 5):

Aspect	Score (out of 5)
Relevance	1.84
Sincerity	2.56
Understandability	2.86
Fluency	3.24
Personalization	1.26

Table 2: Human Evaluation Scores on Generated Responses

4.5 Streamlit Application

To make the system accessible and user-friendly, we built an interactive frontend using **Streamlit**. The interface allows users to input their message, and the system detects the underlying emotion (e.g., *fear, joy, sadness*). Based on the detected emotion and retrieved memory context, the model generates empathetic responses.

A unique aspect of this interface is the **Style Controls** panel on the left, where users can dynamically customize:

- **Formality** (e.g., formal vs. casual)
- **Length** of the response (e.g., short, medium, long)
- **Personality traits** (e.g., warm, funny, assertive)
- **Greeting** prefix
- **Top-k retrieval slider** for RAG document grounding

Users are then presented with multiple generated response options and can select their preferred one. The final assistant message is rendered in the chatbox, simulating real-world AAC usage. This modularity offers both emotional nuance and user control, bringing us closer to a personalized and expressive communication system. The system was tested using the input message: `"I'm scared"`. The model accurately detected the emotion as **fear** and presented two empathetic responses:

- Option 1: *"I'm sure you'll be able to do that!"*
- Option 2: *"I'm sure you'll be able to do that. I hope you get it!"*

The assistant ultimately returned the selected reply in a warm tone, as specified by the user's style settings.

This modular interface offers emotional nuance and user control, enabling more expressive and adaptive communication experiences for AAC users.

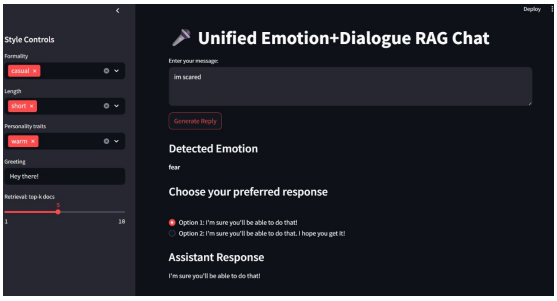

Figure 2: Streamlit Interface: Emotion-Aware Chat with Style Controls and Retrieval Options

5. Conclusion

This project presents a complete end-to-end system to enhance Augmentative and Alternative Communication (AAC) using a combination of fine-tuned language models and retrieval-based personalization. By leveraging the `t5-small` model fine-tuned on emotionally rich datasets like EmpatheticDialogues and GoEmotions, and integrating Retrieval-Augmented Generation (RAG) via FAISS and `all-MiniLM-L6-v2`, the system generates contextually relevant, emotionally aligned responses. A Streamlit-based interface adds interactivity and user control through style preferences, emotion detection, and multiple response choices. Evaluations show promising results, with high semantic similarity (BERTScore $\sim$0.85) and satisfactory fluency ratings from human evaluators. While personalization remains an area for growth, this work demonstrates a scalable, modular approach to building emotionally intelligent AAC tools with real-world usability and impact.

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